Inequalities and Graphs and Static Analysis; Oh My!

Kenny Ballou

California State University San Marcos

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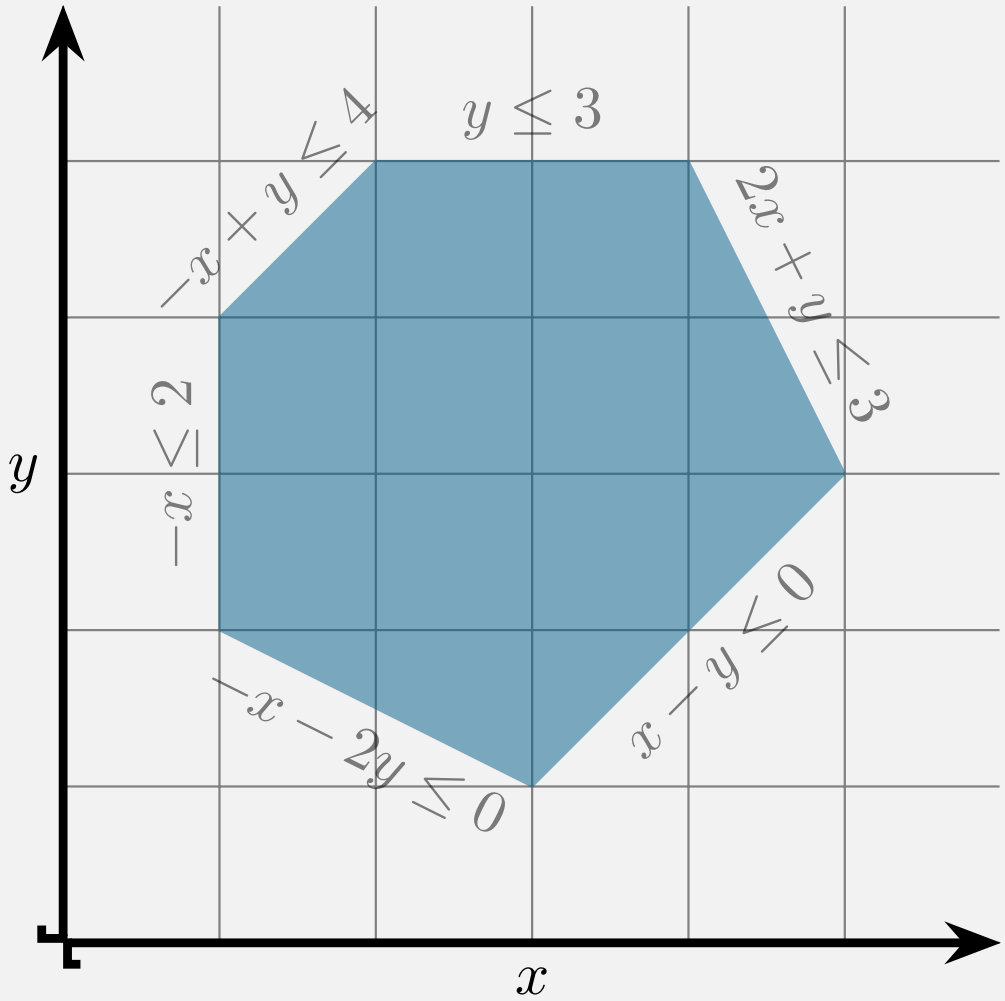
Outline

1. Inequalities
2. Graphs
3. Static Program Analysis

Systems of inequalities can represent available system behavior

$$
\begin{aligned}
-x &\leq 1 \\
-x + y &\leq 4 \\
y &\leq 3 \\
2x + y &\leq 3 \\
x - y &\leq 0 \\
-x - 2y &\leq 0
\end{aligned}
$$

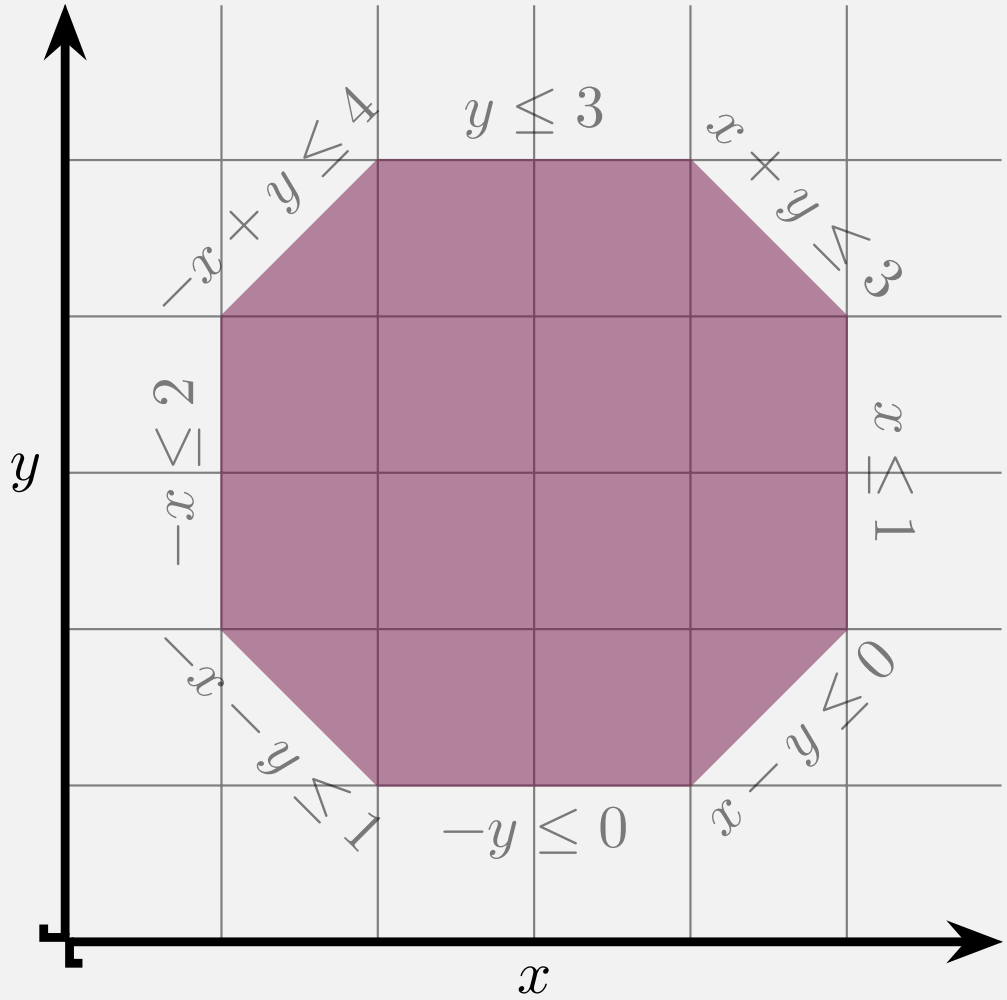
Systems of inequalities can represent available system behavior



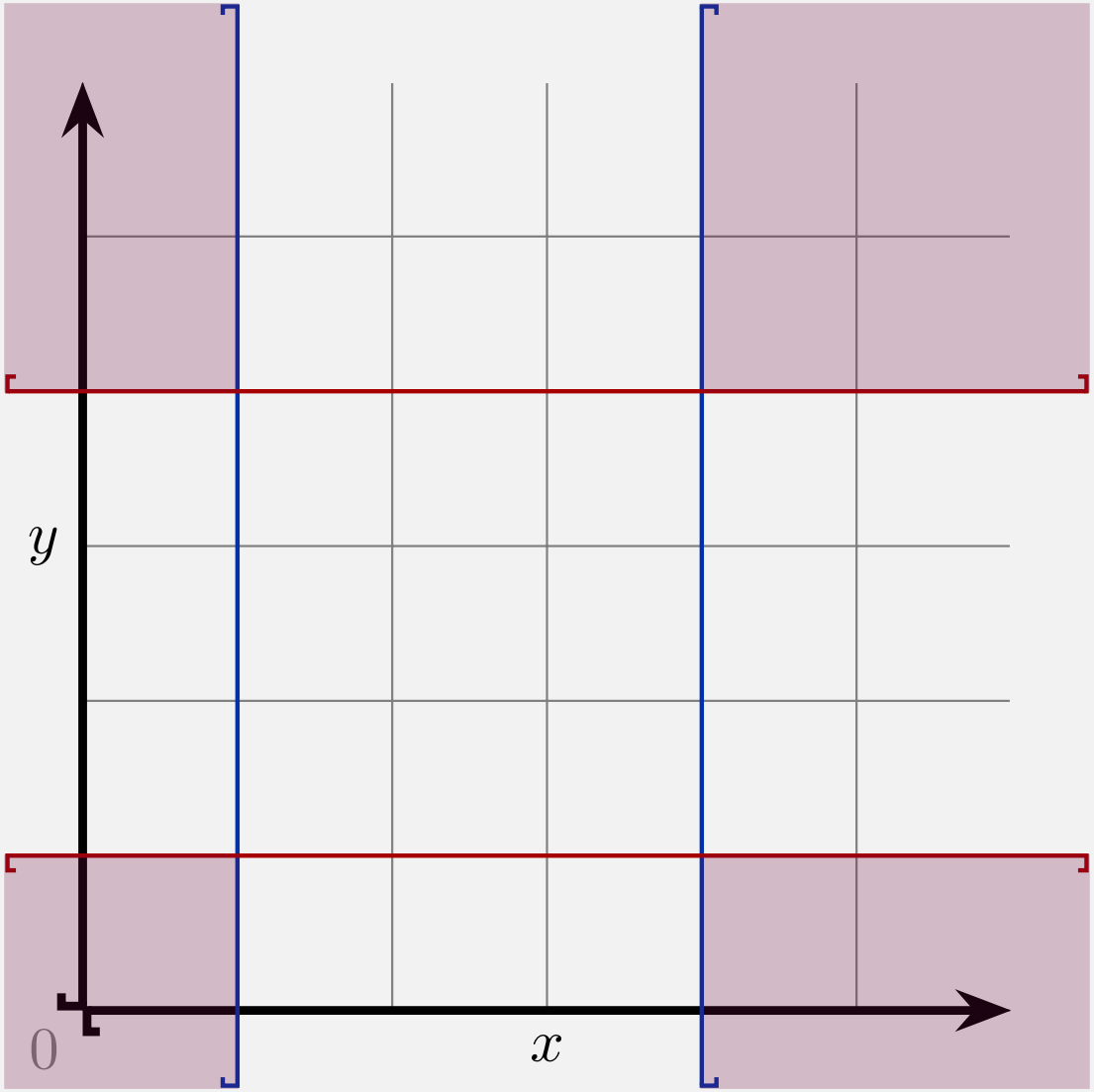
Limiting the form of the inequalities changes the fidelity of our regions

$$
\begin{aligned}
-x &\leq 2 \\
-x + y &\leq 4 \\
y &\leq 3 \\
x &\leq 1 \\
x - y &\leq 0 \\
-y &\leq 0 \\
-x - y &\leq 1
\end{aligned}
$$

Limiting the form of the inequalities changes the fidelity of our regions



Checking inequality plots is tedious



So, we can think real hard and "decide"

$$x \leq 1 \wedge$$
$$x \geq 4 \wedge$$
$$y \leq 1 \wedge$$
$$y \geq 4$$

So, we can think real hard and "decide"



$$x \leq 1 \wedge$$
$$x \geq 4 \wedge$$
$$y \leq 1 \wedge$$
$$y \geq 4$$

But as the systems become larger, what to do?

$$u - v \leq \alpha$$
$$v - w \leq \beta$$
$$x - u \leq \gamma$$
$$w - x \leq \delta$$
$$v - y \leq \epsilon$$
$$z - x \leq \zeta$$
$$r - u \leq \eta$$

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And more complicated???

$$x - y \leq \alpha$$
$$y + z \leq \beta$$
$$w - y \leq \gamma$$
$$u + w \leq \delta$$
$$w - r \leq \epsilon$$
$$z + u \leq \zeta$$

And more complicated???



$$x - y \leq \alpha$$
$$y + z \leq \beta$$
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Let's talk about graphs...

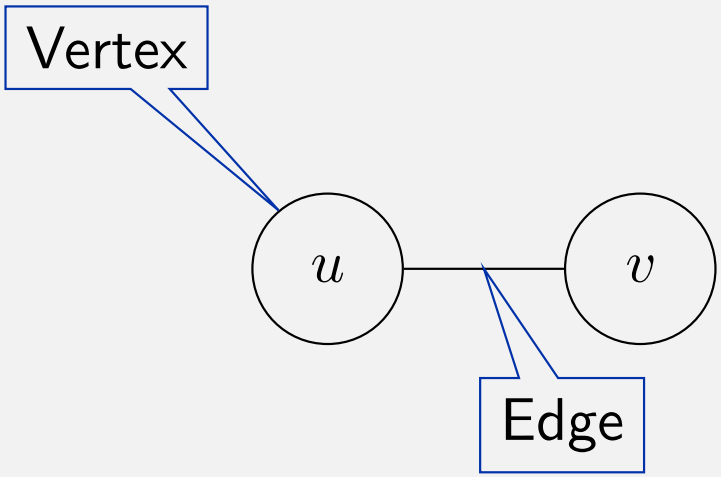
Definition

Graph, G, is a pair (V, E) where V is a set of vertices, and E is a set of pairs between vertices, i.e., $E \subseteq V \times V$.

Let's talk about graphs...

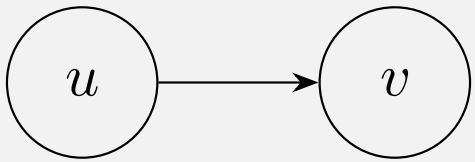
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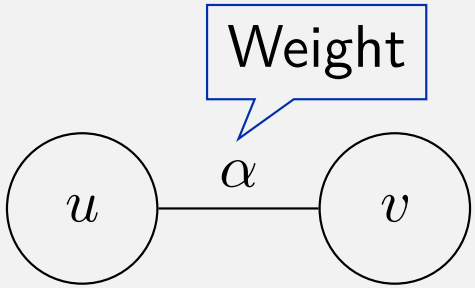
Graph edges can communicate direction

so called, *directed* graphs



Graph edges can carry weights as well

so called, *weighted* graphs



We like graphs because they can represent many real-world problems

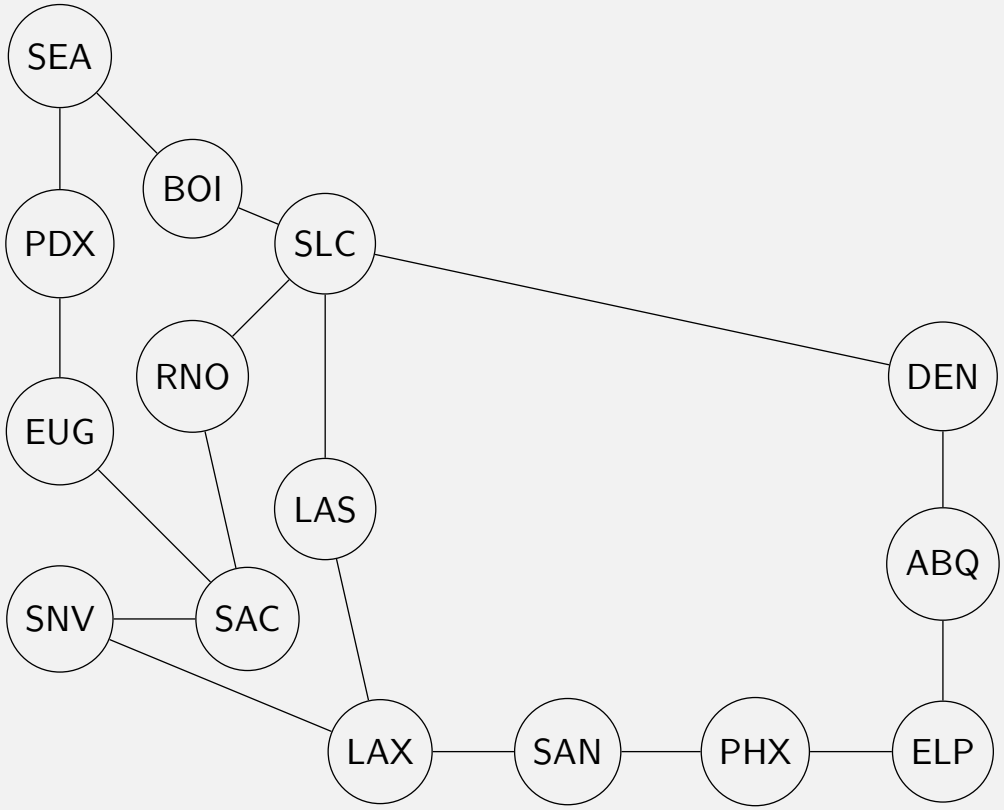
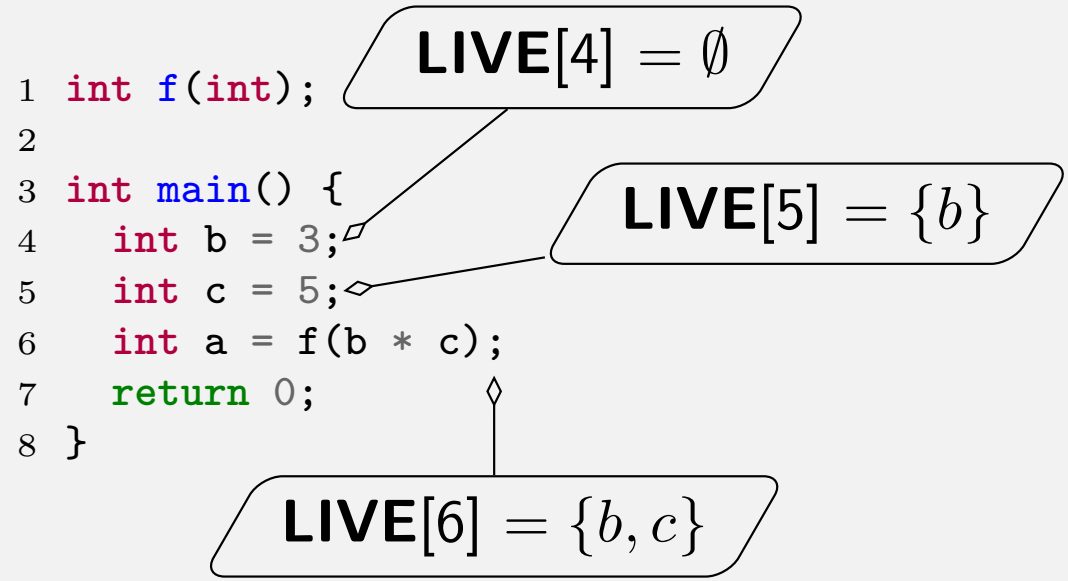


Figure: Internet backbone connections in the south western US

Static analysis computes information about programs

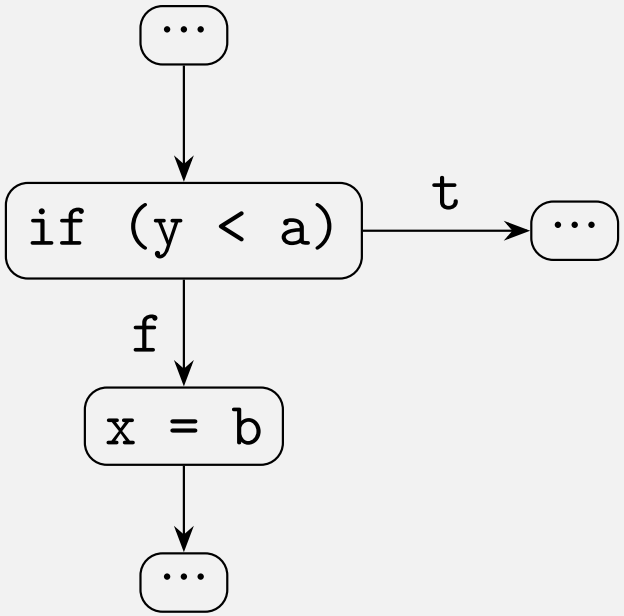
```
1 int f(int);
2
3 int main() {
4   int b = 3;
5   int c = 5;
6   int a = f(b * c);
7   return 0;
8 }
```

Static analysis computes information about programs



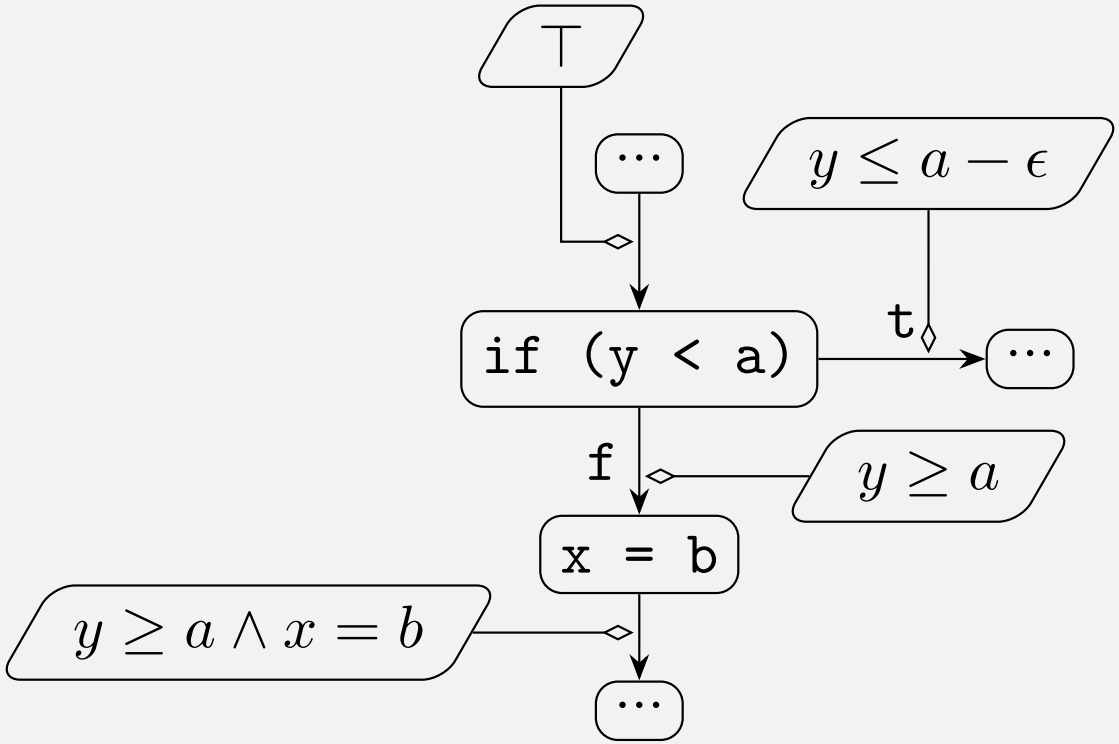
Static analysis *automatically* computes information about programs

```
1 // ...
2 if (y >= a) {
3   x = b;
4   // ...
5 } else {
6   // ...
7 }
```



Static analysis *automatically* computes information about programs

```
1 // ...
2 if (y >= a) {
3   x = b;
4   // ...
5 } else {
6   // ...
7 }
```

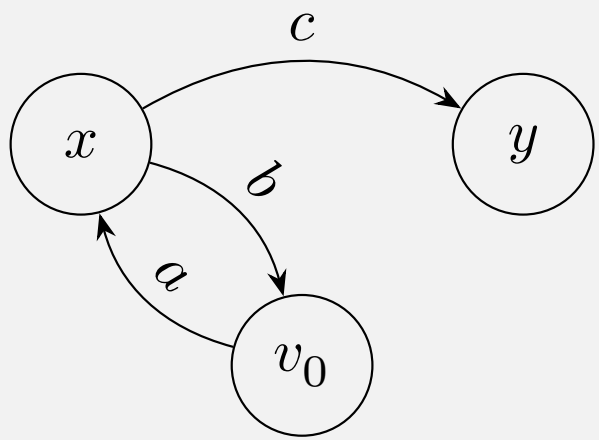


Returning to our constraint systems

Zones Abstract Domain

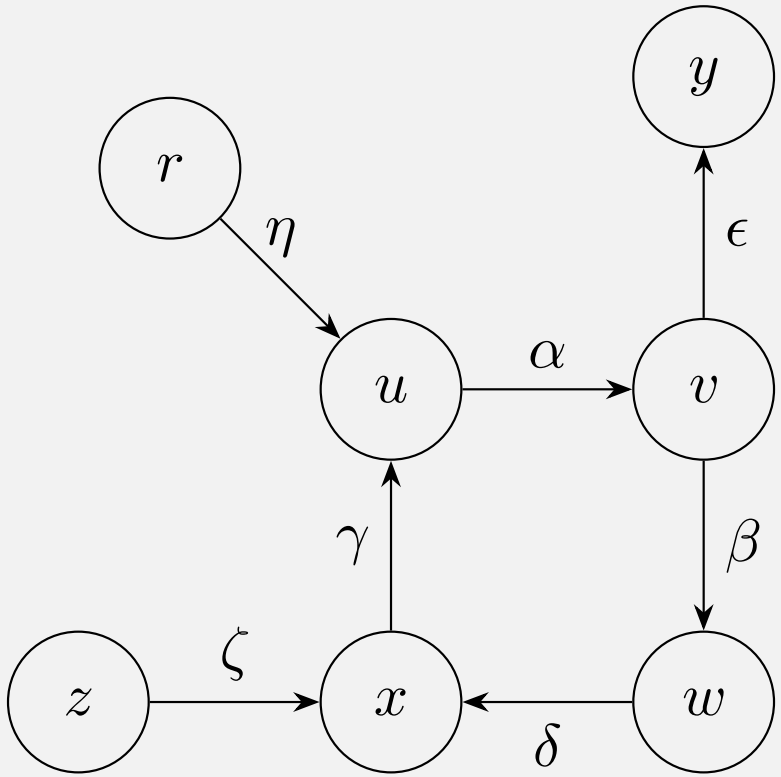
- Encode the strict unit difference between two variables.
- Interval-values use a special variable, $v_0 = 0$

$$x - y \leq c \rightsquigarrow \overrightarrow{xy} = c$$
$$v_0 - x \leq a \rightsquigarrow \overrightarrow{v_0 x} = a$$
$$x - v_0 \leq b \rightsquigarrow \overrightarrow{x v_0} = b$$

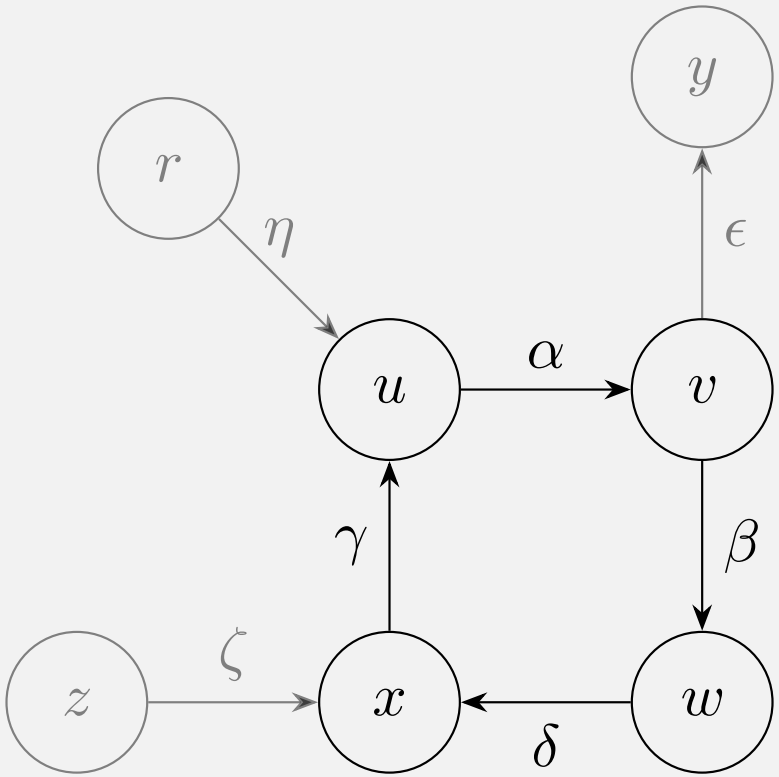


Recall one of our "complicated" systems

$$
\begin{aligned}
u - v &\leq \alpha \\
v - w &\leq \beta \\
x - u &\leq \gamma \\
w - x &\leq \delta \\
v - y &\leq \epsilon \\
z - x &\leq \zeta \\
r - u &\leq \eta
\end{aligned}
$$

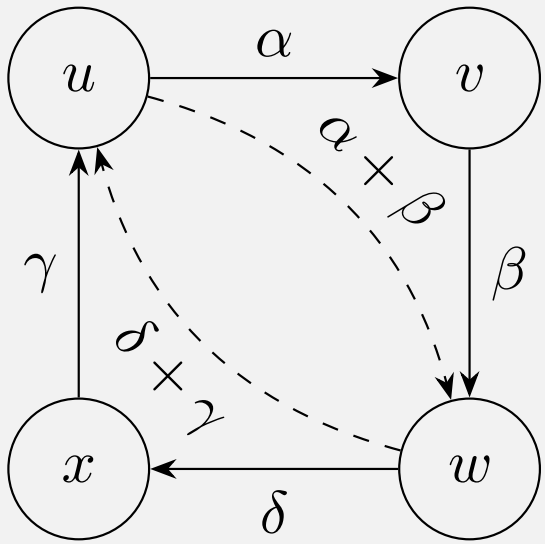


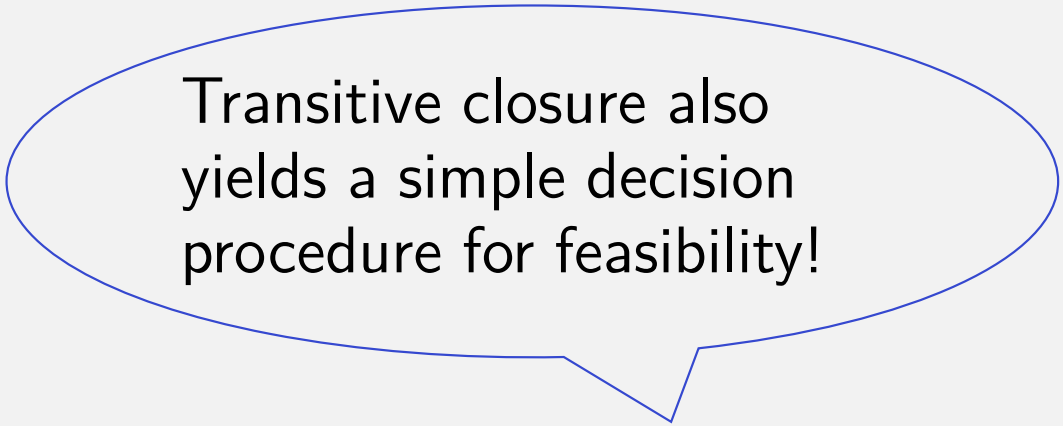
We knew before deciding feasibility required knowing all values



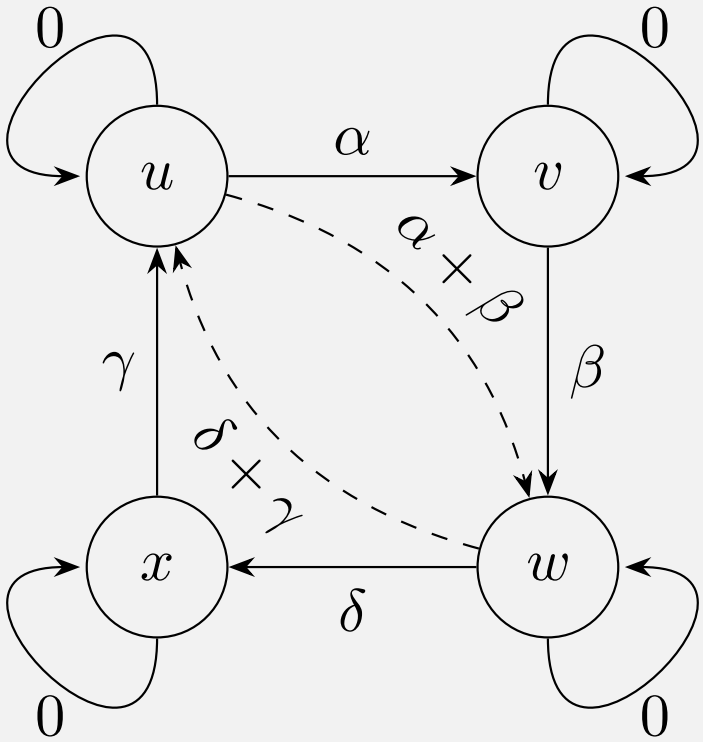
We can also learn new constraints from the *transitive closure*

```
1  void floywdwarshal(int N, int[N][N] mat) {
2    for (int k = 0; k < N; k++) {
3      for (int i = 0; i < N; i++) {
4        for (int j = 0; j < N; j++) {
5          int ikj = mat[i][k] + mat[k][j];
6          if (mat[i][j] >= ikj) {
7            mat[i][j] = ikj;
8          }
9        }
10     }
11   }
12 }
```




Transitive closure also
yields a simple decision
procedure for feasibility!

Checking negative cycles provides a simple decider for feasibility



Value Analysis Demo

Thank You!

Questions?

References I

Glossary I